

for smart camera functions. Here are several capabilities that have become faster and more accurate with Server-on-Edge architecture:

- **Real-time object detection.** Instantly identifies threats
- **Facial recognition.** Matches faces against databases for access control
- **Anomaly detection.** Spots unusual activities to help prevent incidents
- **License plate recognition.** Tracks vehicle movement
- **Crowd analysis.** Supports public safety in high-density areas
- **Behavioural understanding.** Recognises actions such as loitering, tailgating, or sudden crowd formation
- **Dynamic context recognition.** Reads and interprets numbers or other information contextually

AI security cameras now actively analyse video feeds in real-time, using machine learning to distinguish between people, vehicles, and animals. This improves accuracy and reduces false alarms over time. Features such as custom object detection—as seen in solutions like Arlo Secure 5—and LPR systems add monitoring precision. These cameras can also interpret behavioural patterns to flag anomalies and integrate with broader security systems for automated responses, such as triggering alarms or locking doors. The result is more personalised, efficient, and proactive security solutions.

The power of numeric recognition

Number recognition is not new—optical character recognition has been around for decades. But AI is supercharging it. Cameras can now recognise handwritten or digital numbers in real time, even in poor lighting, odd angles, or through obstructions like glass or mesh.

ADVANTAGES AND CHALLENGES OF SERVER-ON-EDGE AI

Advantages	Challenges
Lower latency. AI servers process data locally, eliminating delays caused by cloud transmission	Higher initial investment. Requires robust on-site hardware
Enhanced privacy. Sensitive data remains on-premises, reducing cybersecurity risks	Regular maintenance. AI models need continuous updating to maintain accuracy
Cost-effectiveness. Avoids ongoing cloud subscription fees for video analysis	Power consumption. Persistent AI inference demands significant energy
Scalability. AI models can be tailored to specific security needs	
Reliability. Functions even during internet outages	

This means security cameras can now:

- Track delivery IDs or locker codes as they are entered or displayed
- Read metre readings remotely (useful for utilities)
- Identify specific tags on uniforms, vehicles, or tools in industrial environments
- Monitor prices or time displays in stores for compliance or dynamic pricing alerts
- Count inventory by scanning numbered packaging

All this is achieved with minimal human involvement. While the increase in speed or accuracy may seem insignificant at first, it becomes a game changer when a shoplifter is recognised while still in the showroom—or shortly after they have left. Even more critically, timely alerts can help prevent tragedies such as stampedes or riots.

Anything new?

In the context of security cameras, Gen AI means that the camera is no longer just a passive observer. It becomes an active participant in security, learning from past incidents to predict and prevent future ones. This is particularly useful in surveillance, where predicting suspicious behaviour or identifying hidden threats can be as important as detecting them in real time. AI models, for instance, can be trained to identify unusual movements—such as someone loitering in an area for too long or entering restricted zones. Over time, these systems

refine their understanding of behavioural patterns, reducing false alarms and improving accuracy.

The Dahua DeepSense face detection and recognition network cameras integrate cutting-edge AI technology to deliver accurate and real-time face detection, tracking, and recognition. Powered by deep learning algorithms, these cameras can identify multiple faces simultaneously, even in challenging lighting or crowded environments. Key features include real-time alerts, facial attributes analysis (such as age, gender, and emotion), and efficient database management for secure access control. Ideal for smart surveillance and business intelligence, DeepSense cameras enhance security, streamline operations, and support intelligent decision-making.

The Arlo 5S camera offers AI-powered features that strengthen home security through intelligent detection and real-time alerts. It uses smart object recognition to differentiate between people, animals, vehicles, and packages, thereby reducing false alarms. The camera includes activity zone customisation and learns user preferences over time to improve notification accuracy. With AI-driven motion detection and cloud-based learning, the Arlo 5S delivers responsive and reliable home surveillance.

Artificial intelligence is revolutionising security cameras by enabling them to learn from their environment, leading to major